

About K-NET ASCII format

The K-NET ASCII format is described below with an actual data file.

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-----1-----2-----3-----4-----5-----6-----7-----8
Origin Time      2006/01/18 23:25:00<LF>      (1) Event origin time
Lat.             37.798<LF>                   (2) Event latitude
Lon.            142.200<LF>                   (3) Event longitude
Depth. (km)     36<LF>                       (4) Event depth
Mag.            5.7<LF>                       (5) Magnitude
Station Code    MYG002<LF>                   (6) Site code
Station Lat.    38.7262<LF>                   (7) Site latitude
Station Lon.    141.5109<LF>                   (8) Site longitude
Station Height(m) 79<LF>                     (9) Site altitude
Record Time     2006/01/18 23:25:46<LF>       (10) Recording start time
Sampling Freq(Hz) 100Hz<LF>                 (11) Sampling frequency
Duration Time(s) 108<LF>                     (12) Recording duration time
Dir.            N-S<LF>                       (13) Channel number
Scale Factor    3920(gal)/6182761<LF>         (14) Scale factor
Max Acc. (gal)  48.060<LF>                   (15) Maximum acceleration
Last Correction  2006/01/18 23:25:31<LF>       (16) Time of last correction
Memo.           <LF>                         (17) Comments
      8009      9006      7998      8001      8009      8007      8002      8005<LF>
      8006      8003      8003      8005      8005      8003      8001      8004<LF>
      8008      8004      7998      8002      8009      8005      8001      8002<LF>
      :
      :
      :
      8005      8003      8000      8004      8010      8008      8001      8001<LF>
      8006      8006      8000      8000      8005      8009      8004      7999<LF>
      8003      8007      8004      8000      8000      8005      8009      8005<LF>
[EOF]
-----1-----2-----3-----4-----5-----6-----7-----8

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(Line 9th) Site altitude

The value of -9999 indicates that no site altitude data are available in database.

(Line 10th) Recording start time

This time includes the effect of 15-second trigger delay in the data logger. The true start time of the time series is obtained by subtracting 15 seconds from this "Recording start time."

(Line 13th) Channel number

The strong-motion data for K-NET sites have three data channels that correspond to three components of a seismograph. Relationship between the channels and the extension of the files are as follows.

Channel number	Extension of file
N-S	NS
E-W	EW
U-D	UD

The strong-motion data for KiK-net sites have six channel numbers; the first three correspond to three components of a borehole seismograph and the other three correspond to those of a surface seismograph (see table below). Please note that channels 1 and 2 may be rotated from NS and EW directions due to the difficulty in adjusting the azimuth of sensors in the borehole. Orientations of Hi-net sensor, which are considered to be the same as those of KiK-net sensor installed in the same container, are available from [the Hi-net website](#) (in Japanese) and serve as a reference for KiK-net data.

Channel number	Component	Location	Extension of file
1	NS	Borehole	NS1
2	EW	Borehole	EW1
3	UD	Borehole	UD1
4	NS	Surface	NS2
5	EW	Surface	EW2
6	UD	Surface	UD2

(Line 15th) Maximum acceleration

This value is obtained after the offset of data or the average value of all the data is subtracted.

Strong-motion data begin at line 18th.

Raw data of the seismograph are written as 7-digit integer with sign in eight columns per line. The acceleration time series is obtained by multiplying these data by the scale factor written in line 14th. As the data are provided without corrections, they may contain offsets.